

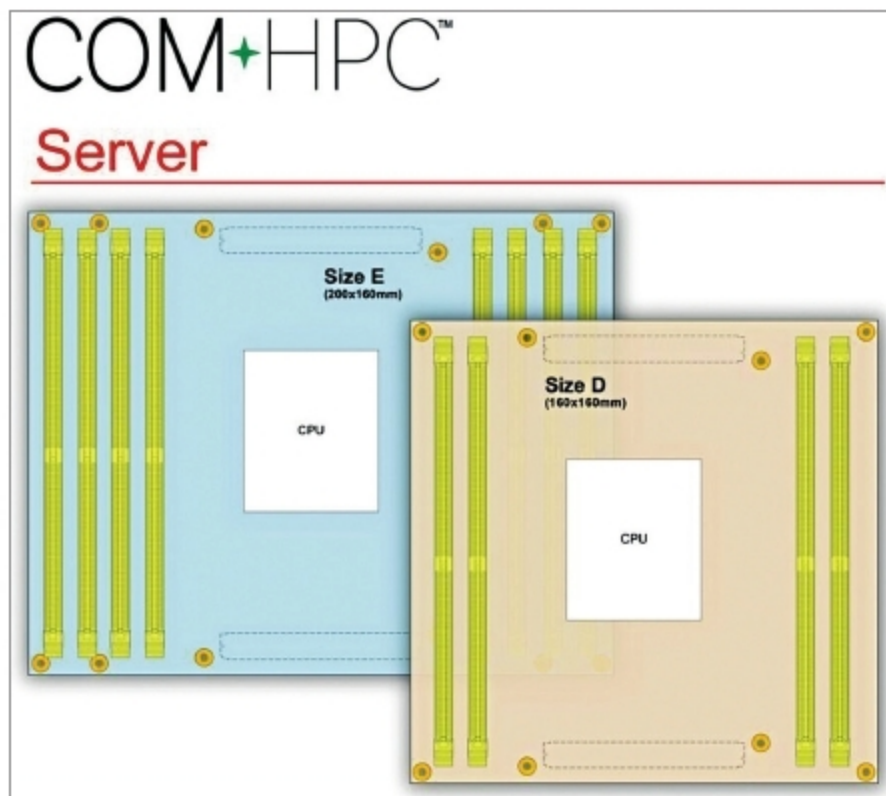


Fig. 3: There are two different COM-HPC server form factors that can host either 4 or 8 SO-DIMM sockets

mising the price/performance ratio for the application. This is a particular advantage for real-time applications where it's not an option to simply scale the load balancing regardless of the real physical resources. Rather it must be possible to allocate real resources in a dedicated way.

When performance upgrades are implemented on the basis of existing system design, the costs for moving to the next generation can be reduced to about fifty per cent of the initial investment because, in majority of the cases, it is enough to simply replace the processor module. This significantly reduces the total cost of ownership (TCO) and accelerates return on investment (ROI). Manufacturer-independent standardisation also ensures second-source strategies, competitive prices, and high long-term availability. Finally, server-on-modules is also more sustainable, ultimately contributing to a lower CO₂ footprint of server technology in production, which—despite all the energy consumed in continuous operation—significantly lowers the environmental impact.

Advanced server management interfaces

Another advantage of the new specification is the availability of a dedicated remote management interface. This interface is currently being defined by the PICMG Remote Management Subcommittee. The aim is to make

(RAMS) as is common in servers. For individual needs, this function can be expanded via the board management controller to be implemented on the carrier board. This provides OEMs with a uniform basis for remote management that can be adapted to specific requirements.

Unbeatable custom interface design and performance upgrades

All in all, COM-HPC server offers many advantages. Whenever edge servers need a very specific mix of interfaces, server-on-modules is the best option. In such cases, a carrier board design can be implemented much faster and more cost-effectively than a full-custom design. Up to eighty per cent of the NRE costs, sometimes even more, can be saved this way. What's more?

COM-HPC server modules are also extremely likely to find widespread use in standard industrial servers as the cost advantages of performance scaling for the next generation are unmatched. The use of modules will further be strengthened by the fact that server performance will increasingly be provided as-a-service so that the investment will be borne by the service provider instead of the user.

COM-HPC: One standard, three application fields

The PICMG COM-HPC standard specifies overall three variants: First, the

part of the function set specified in the intelligent platform management interface (IPMI) available for remote edge server module management. Similar to the slave function, COM-HPC will also provide extended remote management functions to communicate with the modules. This feature guarantees OEMs and users the same reliability, availability, maintainability, and security

so-called COM-HPC server modules, which are positioned above the COM Express Type 7 headless specification, and second the COM-HPC client modules, which are traded as successors of the COM Express Type 6 specification.

A third application field for the specification is FPGA/GPGPU modules for which suitable master/slave functions have been reserved in the standard. While the COM-HPC server and FPGA/GPGPU modules are mainly perceived as innovations, COM-HPC client modules give rise to concerns about the disruptive effect on the popular COM Express Type 6.

Users of these modules are therefore a little more skeptical about the COM-HPC standard. They want to protect existing COM Express investments and may ask themselves: How long will COM Express be offered, and do I have to switch to COM-HPC now? What are the advantages for my customers? They primarily want to know what the COM-HPC client modules have to offer and how they differ from COM Express.

When referring to computer-on-modules instead of client modules, users quickly grasp the new concept. They are familiar with the advantages of these products since the ETX computer-on-module standard was published at the beginning of the millennium.

COM Express computer-on-modules will never disappear quickly as thousands of OEMs use them in their industrial applications, relying on the long-term availability promises of embedded processor and embedded form factor standards. So, you can safely stick with your COM Express designs. There is definitely no need for change as long as the given interface specification suits your requirements.

On the other side, moving from COM Express to COM-HPC designs is definitely recommended for those developers who wish to benefit from the new features such as PCIe Gen 4.0, more than 32 PCIe lanes for massive I/Os, four video interfaces, USB 4.0, 25-Gbit Ethernet, and advanced remote management.

But unless you fall into this group, there is no reason for a rushed change. So, relax and continue to rely on the established open standards. **EFY**